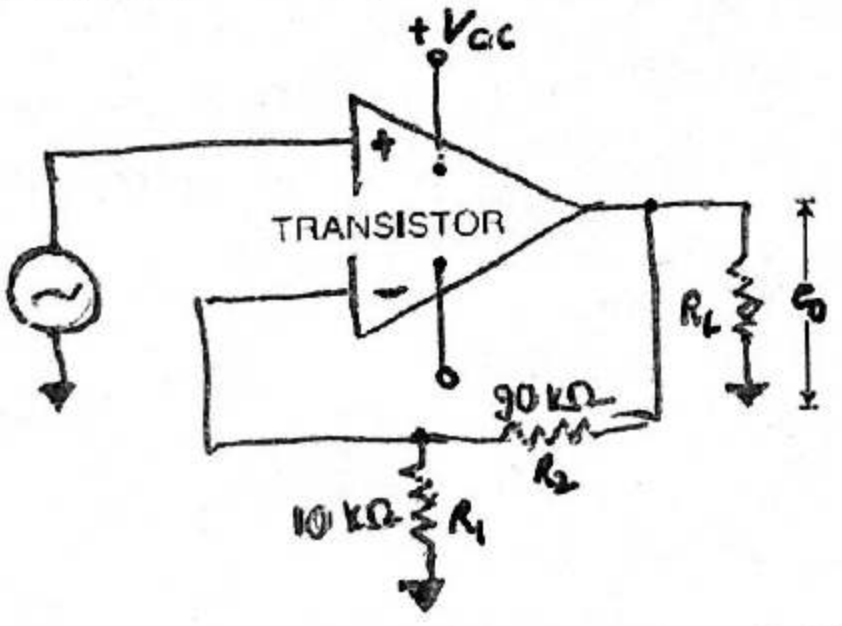
Course Name: B. Tech.-ECE, ECE-AI, Semester: 3rd
(October, 2025) OFF LINE mode

Subject Code: BEC 204	Subject: Advanced Electronic Workshop
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) What is distortion in amplifiers? Explain with example.		CO_1
	b) An amplifier has at its input a signal power of 100 µW and a noise power of 1µW. The amplifier has a power gain of 20 dB. The noise contribution by the amplifier is 100 µW. Find (i) The input S/N ratio (ii) output S/N ratio (iii) noise power factor.		CO_2
	c) An amplifier has an input power of 2µW and a power gain of 30dB. Calculate the output power.		CO_1
	d) Compare positive feedback and negative feedback in amplifiers.		CO_2
	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) In an amplifier, the maximum voltage gain is 2000 and occurs at 2 kHz. It falls to 1414 at 10 kHz and 50 Hz. Find : (i) Bandwidth (ii) Lower cut-off frequency (iii) Upper cut-off frequency (iv) Draw and explain its frequency response plot.		CO_1
	b) Explain the working of RC-coupled amplifier. What are the components of this circuit that affect the bandwidth?		CO_2
	c) Explain the block diagram of multistage amplifier.		CO_1
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Determine trans-conductance and Trans-resistance Amplifier with suitable diagrams.		CO_1
	b) Figure shows the circuit of a negative voltage feedback amplifier. If without feedback, Av = 10,000, Zin = 10 kΩ, Zout = 100 Ω, find: (i) feedback fraction (ii) gain with feedback (iii) input impedance with feedback (iv) output impedance with feedback.  		CO_2
	c) Explain input-output resistances of current series feedback amplifier with block diagram.		CO_2